Figure 1: Login prompt shown by calling `dbloginform.html`

```
<html><head><title>Login</title></head>
<body>
<form method="post" action="dblogin.php">
<center>
<table border=0>
<tr>
<td><label for='username'>Username:</label></td>
<td><input type="text" id="username" name="username" maxlength="50"
/></td>
</tr>
<tr>
<td><label for='passwd'>Password:</label></td>
<td><input type="password" id="passwd" name="passwd" /></td>
</tr>
</table><br />
<input type="submit" value="Log in" />
</form>
</body></html>
```

The resulting login screen is shown in Figure 1.

Logging in

The login process is handled by the script `dblogin.php`, which takes the user name and password and checks for the validity of the pair. After doing so, a session ID is generated, which needs to be a unique number. In the present case, the `microtime()` function with 'true' as an argument was called and the result was printed into a string called `$t`. A `while` loop was run until the session ID got inserted into the database. The login screen is shown in Figure 2 and the login script is provided below:

```
<?php
/*dblogin.php*/
$user;
$session_id;
function dblogin($u, $p)
{
    global $user, $session_id;
```

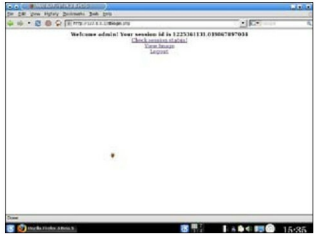


Figure 2: Login screen for server-side session management

```
$con = mysql_connect('localhost','user','pass') or die(mysql_error());
mysql_select_db('session',$con) or die(mysql_error());

$res=mysql_query("select id, decode(pass,'session') as pass from user where
name='".$addslashes($u)."'"); or die(mysql_error());
if(mysql_num_rows($res) != 1) {
    printf("<center><h1>User %s not found</h1><br /><a href='
'dbloginform.html'>Go to login page</a></center>", $username);
    return false;
}
if(mysql_result($res,0,"pass") != $p) {
    printf("<center><h1>Login attempt rejected for %s</h1><br /><a
href='dbloginform.html'>Go to login page</a></center>", $username);
    return false;
}

$id = mysql_result($res,0,"id");
$t = sprintf("%.12f",microtime(true));
$done = false;
while($done) {
    $done = true;
    mysql_query("insert into session_log values ($t,$id,'_
SERVER(REMOTE_ADDR)'.1.now(),now())"); or $done=false;
    if($done)
        $t = sprintf("%.12f",microtime(true));
}
$user=$u;
$session_id=$t;
mysql_close($con);
return true;
}

if(dblogin($_POST['username'], $_POST['passwd']))
    printf("<center><b>Welcome %s! Your session id is %s</
b><br /><n'>",$user, $session_id);
    printf("<a href='dbloginstatus.php'>Check session status</
a><br /><a href='dbprotectedimage.php'>View Image</a><br /><a
href='dblogout.php'>Logout</a></center>");
?>
```